AutoMetric AI

AI Assisted Car Valuation and Maintenance

ABSTRACT



Buying or selling a used car often begins with one question: what is a reasonable asking price? AutoMetric AI uses vehicle listings from Pakistan to estimate prices from a car's specifications and mileage. Available on Android and web, it gives buyers and sellers a useful starting point for comparison and negotiation. Owners can also keep service records and plan upcoming maintenance in the same app.

PROBLEM STATEMENT

Similar-looking cars can carry very different asking prices. Checking listings manually takes time, while differences in age, mileage and variant make comparisons difficult. Service records and maintenance deadlines are also easy to lose track of.



AIM AND OBJECTIVES



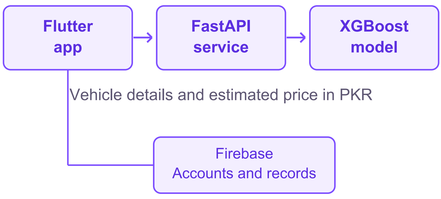
Aim: Provide a locally relevant price reference and a simple way to manage vehicle upkeep.

Objectives

- Estimate prices using vehicle details and local listing data.
- Save valuations for later comparison.
- Organize service records and upcoming maintenance.

METHODOLOGY

- **Collect:** Gather used car listings from PakWheels.
- **Prepare:** Clean and standardize vehicle data.
- **Develop:** Compare models and select XGBoost.
- **Connect:** Package the prediction API with Docker.
- **Build:** Integrate Flutter with Firebase records.

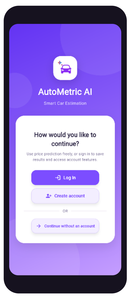


RESULTS

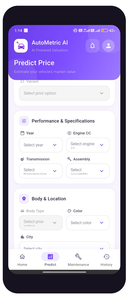
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vehicle records prepared for model development

- Price estimation on Android and web.
- Saved valuations and a garage for multiple cars.
- Service history and reminders based on dates and user-updated mileage.



Welcome screen



Price prediction

Estimates reflect advertised prices and do not replace a vehicle inspection.

TECHNOLOGIES USED



Flutter · Python · XGBoost · Firebase · FastAPI · Docker

SUMMARY

AutoMetric AI gives buyers and sellers a data-based starting point for price discussions, while helping owners keep vehicle care organized.

Demo QR

PROJECT DEMO

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